

ROYAN SABILA ROSYAD WAHYUDI

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Summary

AI Engineer specializing in **production LLM applications, RAG pipelines, and agentic workflows**. Shipped multi-agent systems (Mastra, LangChain, LangGraph, Deep Agents) at Zurich Insurance and Indonesian government scale, integrating Pinecone vector stores, Tavily web retrieval, and LangSmith evaluation to move GenAI from prototype to production. Reduced manual workflows by **60–75%** through agent orchestration, prompt engineering, and input-validation guardrails. Informatics Engineering graduate (Cumlaude, 3.83/4.00, UIN Syarif Hidayatullah Jakarta) combining LLM app development, vector retrieval, FastAPI serving, and containerized CI/CD on GCP and Azure.

Work Experience

Associate AI Engineer – Straitpoint Indonesia
Fulltime

June 2026 – Present

- Built and productionized an **LLM-enhanced personalized recommender system** that combines child-profile metadata, free-text preference signals, embedding models, and vector store retrieval to deliver relevant content for new or interaction-sparse users, helping mitigate the **user cold-start problem** without relying solely on behavioral history.
- Developed an end-to-end AI customer-service dashboard using LangChain and LangGraph, integrating RAG, tool calling, conversation routing, guardrails, and human handoff workflows.
- Build internal tools using Mastra agents with OpenWebUI to provide an internal conversational interface, exploring email processing, workflow automation, and task scheduling use cases.

Data Scientist (AI Engineer Focused) – Zurich Insurance Indonesia
Apprenticeship

Nov 2025 – June 2026

- Built a **multi-agent LangGraph system** with Tavily retrieval and a Reflex UI that autonomously researches companies, fills profile gaps, and surfaces evidence for underwriting instrumented with **LangSmith tracing** and input-validation guardrails to prevent hallucinations on real insurance data.
- Shipped a **Deep Agents pipeline** automating research workflows and intelligent Excel claim processing; standardized schemas, derived actuarial columns, and **reduced manual workflow by 60%**.
- Designed **prompt templates, tool-calling contracts, and evaluation rubrics** to keep agent outputs grounded and reproducible across runs.

AI Automation Engineer – Government of Sukabumi
Freelance

Apr 2025 – Sep 2025

- Designed an end-to-end **LLM-powered sentiment pipeline** using n8n to process high-volume citizen comments across social media, chaining social APIs, sentiment models, and response generation.
- **Reduced manual review time by 75%** and improved response consistency across government communications by enforcing prompt templates and escalation rules.

AI Engineer – Laskar AI
Apprenticeship

Feb 2025 – Aug 2025

- Fine-tuned and evaluated ML and deep learning models on real-world datasets; tracked experiments, versions, and metrics in **MLflow on Azure ML** for reproducibility and collaboration.
- Applied supervised and unsupervised learning end-to-end (preprocessing, training, evaluation, deployment) before promoting models to production.

Education

Universitas Islam Negeri Syarif Hidayatullah Jakarta – Indonesia
Bachelor of Informatics Engineering, 3.83/4.00 (Cumlaude)

Sep 2021 – Aug 2025

- Relevant Coursework: Artificial Intelligence, Data Structures & Algorithms, Database Management, Data Science, Web Development, Project Management.

Projects

RAG Chatbot for Chicken Disease Diagnosis (LangChain + Pinecone) [🔗](#)

- Shipped an end-to-end **RAG application** with LangChain orchestration, Pinecone vector store, and image-classification backbone diagnosing 4 chicken diseases at **95% accuracy**.
- Engineered chunking strategy, embedding selection, and retrieval/reranking so answers are **grounded with citations** and low hallucination on domain data.

AI Company Enrichment (LangGraph Multi-Agent + Tavily) [🔗](#)

- Designed an **agentic workflow in LangGraph** coordinating retrieval, reasoning, and step-by-step enrichment to automate lead research for profiling workflows.
- Integrated **Tavily real-time web search** as a retrieval tool; defined tool contracts, state machines, and fallback prompts so agents recover from missing data.

Student Dropout Risk Prediction (Deployed ML + Dashboard) [🔗](#)

- Built a Gradient Boosting pipeline achieving **90% accuracy and 84% recall**, deployed via Streamlit for direct stakeholder input and instant prediction.
- Shipped a Metabase dashboard that turned model outputs into decision support for university stakeholders.

License & Certifications

Associate AI Engineer for Developers – DataCamp [🔗](#)

Issued Sep 2025 – Expires Sep 2027

Developing Machine Learning Systems in Production – Dicoding Indonesia [🔗](#)

Issued Jun 2025 – Expires Jun 2028

Applied Machine Learning – Dicoding Indonesia [🔗](#)

Issued May 2025 – Expires May 2028

Skills

LLM & GenAI: Retrieval-Augmented Generation (RAG), Prompt Engineering, Chain-of-Thought, Function/Tool Calling, Multi-Agent Orchestration, Pinecone, Reranking, Embedding Models

LLM Frameworks: Mastra Agent, LangChain, LangGraph, LangSmith, Deep Agents, OpenAI Agent SDK, Azure OpenAI API, Anthropic Claude API, n8n.

Data Visualization & BI Tools: Tableau, Looker Studio, Metabase.

Programming & Serving: Python, SQL, FastAPI, REST APIs, Git.

ML/DL: PyTorch, TensorFlow, Scikit-learn, NLP, Classical ML.

MLOps & Deployment: Docker, Kubernetes, CI/CD, MLflow, Prometheus, Grafana, LangSmith observability, Redis caching.

Cloud: GCP (Vertex AI), Azure (Azure ML/OpenAI), AWS.

Soft Skills: Communication, Critical Thinking, Problem Solving, Collaboration, Adaptability.